



**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

# LESSON PLAN

For

## **Certified Lean Six Sigma Green Belt (CLSSGB)**

Course Code: C481

Conducted by

**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

**Version 4**

## DOCUMENT VERSION CONTROL RECORD

| Version Number | Effective Date of Release | Summary of Included Changes                                                                                                                                                                                                                                                                                                                                     | Author         |
|----------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 1              | 1 June 2026               | Initial release - CLSSGB lesson plan.                                                                                                                                                                                                                                                                                                                           | Dr. Alfred Ang |
| 3              | 19 July 2026              | Removed the Practice Exam slide and its asset - there is no Six Sigma practice exam on <a href="https://exams.tertiaryinfotech.com">exams.tertiaryinfotech.com</a> .                                                                                                                                                                                            | Dr. Alfred Ang |
| 4              | 19 July 2026              | Added the SIPOC & Process Map Builder ( <a href="https://alfredang.github.io/sipoc">alfredang.github.io/sipoc</a> ) to the toolkit - wired into Lab 6 (SIPOC build, pain points, 'Check my SIPOC' validation) and Lab 7 (swimlane and handoff table), the Define phase slides, and a new Your Interactive Toolkit slide that introduces all five browser tools. | Dr. Alfred Ang |

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## Course Information

|                          |                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Course Title</b>      | Certified Lean Six Sigma Green Belt (CLSSGB)                                                                     |
| <b>Course Reference</b>  | C481                                                                                                             |
| <b>Training Provider</b> | Tertiary Infotech Academy Pte Ltd (UEN 201200696W)                                                               |
| <b>Duration</b>          | 4 days · 7.5 instructional hours per day (30 hours)                                                              |
| <b>Daily Timing</b>      | 9:30 am – 5:30 pm (30-minute lunch; refreshments taken during activity time)                                     |
| <b>Mode</b>              | Instructor-led, hands-on Lean Six Sigma labs using the Northwind Retail Distribution Centre improvement scenario |
| <b>Trainer</b>           | Dr. Alfred Ang                                                                                                   |

## Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Lead a Lean Six Sigma project — define the problem, scope the work and charter the project (A1, A2).
- LO2: Map and baseline a process using SIPOC, detailed process maps and value stream maps (A2).
- LO3: Build a valid measurement system — sampling, sample size and MSA/Gage R&R — and baseline capability (A4).
- LO4: Analyse process data statistically using Pareto, run charts, hypothesis testing, correlation and regression (A3).
- LO5: Identify and prove root causes of variation using Fishbone, 5 Whys and statistical evidence (A3, K2).
- LO6: Select, risk-assess and pilot improvements using solution selection matrices, FMEA and DOE (A5).
- LO7: Sustain the gain with SPC control charts, process capability and a control plan (A5, K1).

## Course Schedule

### Day 1 — Six Sigma foundations, the Green Belt role and the Define phase

| Time        | Duration | Topic / Activity                                                                                                        |
|-------------|----------|-------------------------------------------------------------------------------------------------------------------------|
| 9:30–10:00  | 30 min   | Welcome, course introduction and ground rules [slides 2–16]                                                             |
| 10:00–11:15 | 75 min   | FOUNDATIONS — What is Quality; Lean vs Six Sigma vs Lean Six Sigma; the five Lean principles; history of Six Sigma; Y = |

|             |        |                                                                                                                                                                                                                        |
|-------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             |        | f(X); classical vs Six Sigma quality [slides 19–57]                                                                                                                                                                    |
| 11:15–12:15 | 60 min | FOUNDATIONS — sigma level and DPMO; the 1.5 sigma shift; Cost of Poor Quality; belt roles and what a Green Belt leads; project selection and the cause/solution matrix; the DMAIC roadmap and tollgates                |
| 12:15–12:45 | 30 min | Lunch break                                                                                                                                                                                                            |
| 12:45–14:15 | 90 min | Hands-on: Lab 1: The Green Belt Role, Project Selection and Business Case; Lab 2: $Y = f(X)$ , Sigma Level, DPMO and the DMAIC Roadmap [slides 58–65]                                                                  |
| 14:15–15:45 | 90 min | DMAIC · DEFINE — Voice of the Customer; affinity diagrams; Kano analysis; CTQ trees and operational definitions; the project charter; problem and goal statements; scope; SIPOC; stakeholders and RACI [slides 66–120] |
| 15:45–17:00 | 75 min | Hands-on: Lab 3: Voice of the Customer, Affinity Diagram and Kano Analysis; Lab 4: CTQ Tree — Translating Customer Needs into Measurable Requirements [slides 121–128]                                                 |
| 17:00–17:30 | 30 min | Day 1 recap and Q&A                                                                                                                                                                                                    |

Total instructional time: 450 minutes (7.5 hours), 9:30am – 5:30pm.

## Day 2 — Measure — process mapping, sampling, MSA and baseline capability

| Time        | Duration | Topic / Activity                                                                                                                            |
|-------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 9:30–9:45   | 15 min   | Day 1 recap and questions                                                                                                                   |
| 9:45–11:00  | 75 min   | Hands-on: Lab 5: Project Charter, Problem Statement, Goal Statement and Scope; Lab 6: SIPOC, Stakeholder Analysis and RACI [slides 129–138] |
| 11:00–12:15 | 75 min   | DMAIC · MEASURE — process mapping symbols                                                                                                   |

|             |         |                                                                                                                                                                                                                               |
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|             |         | and swimlanes; handoff analysis; the eight wastes (DOWNTIME); value-added analysis; value stream mapping; lean metrics and takt time [slides 139–213]                                                                         |
| 12:15–12:45 | 30 min  | Lunch break                                                                                                                                                                                                                   |
| 12:45–14:15 | 90 min  | Hands-on: Lab 7: Detailed Process Mapping and Swimlane Analysis; Lab 8: Value Stream Mapping, Takt Time and the Eight Wastes [slides 214–223]                                                                                 |
| 14:15–15:30 | 75 min  | DMAIC · MEASURE — types of data; data collection plans; sampling techniques; sample size calculation; Measurement System Analysis and Gage R&R; yield, DPU, DPO, DPMO and RTY; descriptive statistics and baseline capability |
| 15:30–17:15 | 105 min | Hands-on: Lab 9: Data Types, Operational Definitions and the Data Collection Plan; Lab 10: Sampling Techniques and Sample Size Calculation; Lab 11: Measurement System Analysis and Gage R&R [slides 224–237]                 |
| 17:15–17:30 | 15 min  | Day 2 recap and Q&A                                                                                                                                                                                                           |

*Total instructional time: 450 minutes (7.5 hours), 9:30am – 5:30pm.*

### **Day 3 — Analyze — root cause, hypothesis testing, correlation and regression**

| <b>Time</b> | <b>Duration</b> | <b>Topic / Activity</b>                                                                                                                                         |
|-------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9:30–9:45   | 15 min          | Day 2 recap and questions                                                                                                                                       |
| 9:45–11:00  | 75 min          | Hands-on: Lab 12: Yield, DPU, DPO, DPMO, RTY and the Hidden Factory; Lab 13: Descriptive Statistics, Normality and Baseline Process Capability [slides 238–248] |
| 11:00–12:15 | 75 min          | DMAIC · ANALYZE — common vs special cause variation; tampering; run                                                                                             |

|             |         |                                                                                                                                                                                                                                         |
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|             |         | charts and the six non-random patterns; the Pareto principle; stratification; boxplots [slides 249–319]                                                                                                                                 |
| 12:15–12:45 | 30 min  | Lunch break                                                                                                                                                                                                                             |
| 12:45–14:00 | 75 min  | Hands-on: Lab 14: Variation, Run Charts and Stability Analysis; Lab 15: Pareto Analysis, Stratification and Boxplots [slides 320–329]                                                                                                   |
| 14:00–15:30 | 90 min  | DMAIC · ANALYZE — Fishbone (5M+E); 5 Whys; multi-voting; hypothesis testing ( $H_0/H_a$ , $\alpha$ , Type I and II errors, test selection, p-value decision rule); correlation and the coefficient of determination; regression         |
| 15:30–17:15 | 105 min | Hands-on: Lab 16: Fishbone, 5 Whys, Multi-Voting and Cause Prioritisation; Lab 17: Hypothesis Testing — Test Selection, p-values and Conclusions; Lab 18: Correlation, Regression and Quantifying the X-Y Relationship [slides 330–347] |
| 17:15–17:30 | 15 min  | Day 3 recap and Q&A                                                                                                                                                                                                                     |

Total instructional time: 450 minutes (7.5 hours), 9:30am – 5:30pm.

## Day 4 — Improve, Control and project closure

| Time        | Duration | Topic / Activity                                                                                                                                                                                                          |
|-------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9:30–9:50   | 20 min   | Day 3 recap and questions                                                                                                                                                                                                 |
| 9:50–11:15  | 85 min   | DMAIC · IMPROVE — solution generation and brainwriting; benchmarking; the solution selection matrix; 5S; poka-yoke; pull, JIT and Heijunka; standard work; FMEA and RPN; Design of Experiments; piloting [slides 348–394] |
| 11:15–13:25 | 130 min  | Hands-on: Lab 19: Solution Generation,                                                                                                                                                                                    |

|             |        |                                                                                                                                                                                                                                                       |
|-------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             |        | Benchmarking and Brainwriting; Lab 20: Lean Countermeasures — 5S, Poka-Yoke, Pull, JIT and Standard Work; Lab 21: Solution Selection Matrix and Cost-Benefit Analysis; Lab 22: FMEA, Risk Priority Numbers, DOE and Piloting [slides 395–416]         |
| 13:25–13:55 | 30 min | Lunch break                                                                                                                                                                                                                                           |
| 13:55–15:20 | 85 min | DMAIC · CONTROL — process capability Cp and Cpk; control limits vs specification limits; SPC and control chart selection; the eight out-of-control rules; the control plan; SOPs; visual management; verifying the gain and handover [slides 417–468] |
| 15:20–16:25 | 65 min | Hands-on: Lab 23: Statistical Process Control — Chart Selection and Control Limits [slides 469–475]                                                                                                                                                   |
| 16:25–16:45 | 20 min | Hands-on: Lab 24: Control Plan, SOP, Visual Management and Response Plan; Lab 25: Verify the Gain, A3 Storyboard, Handover and Project Closure [slides 476–488]                                                                                       |
| 16:45–17:30 | 45 min | Course recap, wrap-up and next steps                                                                                                                                                                                                                  |

*Total instructional time: 450 minutes (7.5 hours), 9:30am – 5:30pm.*

## Lab Reference (aligned to the DMAIC phases)

| DMAIC phase / Topic                                                 | Weighting | Labs                                                |
|---------------------------------------------------------------------|-----------|-----------------------------------------------------|
| <b>FOUNDATIONS: Six Sigma Foundations &amp; The Green Belt Role</b> | 10%       | Lab 1, Lab 2                                        |
| <b>DEFINE: Define — Scope the Problem</b>                           | 18%       | Lab 3, Lab 4, Lab 5, Lab 6                          |
| <b>MEASURE: Measure — Quantify Performance</b>                      | 24%       | Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13 |
| <b>ANALYZE: Analyze — Find</b>                                      | 26%       | Lab 14, Lab 15, Lab 16, Lab                         |

|                                                             |     |                                |
|-------------------------------------------------------------|-----|--------------------------------|
| <b>and Prove the Root Cause</b>                             |     | 17, Lab 18                     |
| <b>IMPROVE: Improve — Select, De-Risk and Pilot the Fix</b> | 12% | Lab 19, Lab 20, Lab 21, Lab 22 |
| <b>CONTROL: Control — Hold the Gain</b>                     | 10% | Lab 23, Lab 24, Lab 25         |